

Novel active force feedback glove and control algorithm for a realistic virtual handshake

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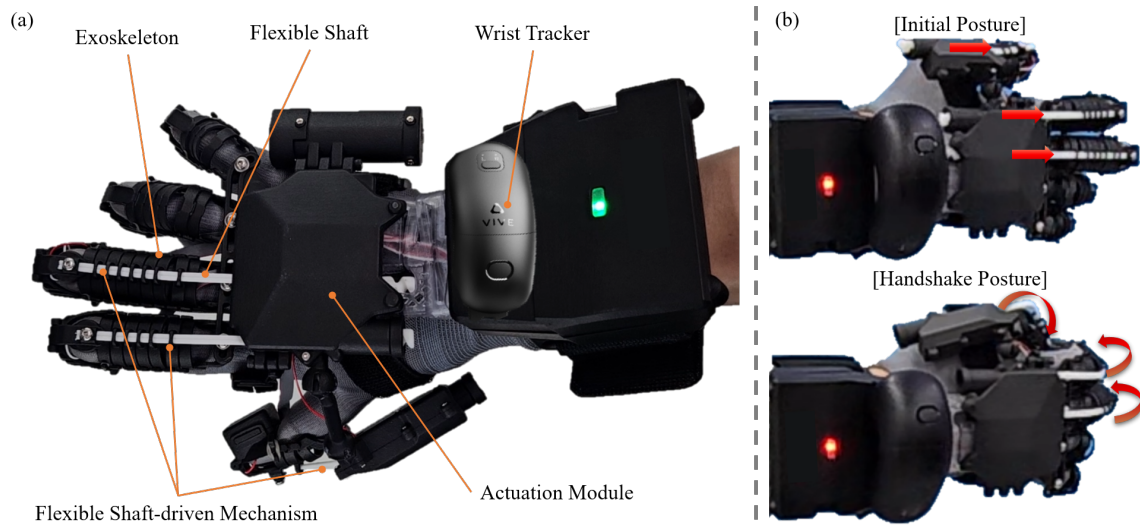


Figure 1: (a) Novel active force feedback glove with an exoskeleton accompanied by a flexible shaft-driven mechanism and (b) Schematic of the handshake movement.

Abstract

Active force feedback is essential for enhancing the sense of presence in eXtended Reality (XR) environments by enabling various virtual physical sensations, such as stopping, rebounding, or impacting. For these requirements, we propose a novel active force feedback glove with an exoskeleton accompanied by a flexible shaft-driven mechanism, configured for the thumb, index, and middle fingers. A timing-based open-loop control algorithm controls this glove, regulating the actuation duration. It simulates a handshake

posture by activating a flexible shaft-driven mechanism. This system, consequently, enables users to experience the realistic sensation of the fingers actively being guided into a handshake, thereby significantly enhancing the sense of presence in XR environments.

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1 Introduction

Haptic gloves [A. Adilkhanov and Kappassov 2022; Manuel Caeiro-Rodríguez and Llamas-Nistal 2021] play a crucial role in delivering the sense of presence of virtual activities in XR environments, but most rely on vibrotactile or passive force feedback, which limits dynamic interactions such as stopping, rebounding, pushing, or impacting. To address these challenges, we propose a novel active

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force feedback glove and its control algorithm. This glove employs an exoskeleton accompanied by a flexible shaft-driven mechanism, achieving a compact and lightweight design. The control algorithm is designed based on a timing-based open-loop control algorithm. By regulating the actuation duration of the mechanism, it produces realistic dynamic interactions in a simple yet effective manner. To validate the feasibility of the system, we implemented a virtual handshake system in an XR environment. We present the active force feedback glove and its control algorithm, successfully guiding users' fingers into a handshake posture, allowing users to experience a realistic virtual handshake in the XR environment.

2 Method

The active force feedback glove consists of a flexible shaft-driven mechanism, an actuation module, and an exoskeleton (Figure 1(a)). The flexible shaft-driven mechanism employs a flexible shaft, one end of which is attached to an actuation module and the other to a thimble. The actuation module consists of a geared motor and a lead screw, allowing control of shaft displacement through the motor speed and actuation duration. The exoskeleton is composed of a series of arched segments forming a mechanical chain. One end is attached to a base structure that anchors the glove to the hand, and the other to the thimble. Above the arched segment is a tunnel structure that guides the direction of the flexible shaft. As the actuation module drives the flexible shaft, the interactive force is transmitted to the fingertip through the flexible shaft-driven mechanism and the exoskeleton. Owing to the shaft's flexibility and the segmented exoskeleton, the active force feedback glove ensures natural finger movement while delivering accurate interactive forces for the finger's flexion/extension.

A timing-based open-loop control algorithm implements realistic virtual handshakes in an XR teleconferencing scenario. To form the handshake posture, the actuation module pushes the fingers in the distal direction through the flexible shaft and the exoskeleton with a force of up to 10 N, bending the finger joints. The lead screw transmission determines the displacement of the flexible shaft using the number of motor revolutions, calculated by the product of the motor speed and actuation duration. Because the motor speed is constant (via a constant 5 V power supply), the shaft's displacement is controlled by only the actuation duration. The preliminary experiment determined that 2 seconds of actuation duration forms a handshake posture. Hence, once the glove enters a predefined collision box in the XR environment, the actuation module is activated in the direction of pushing the shaft for 2 seconds, thereby forming the handshake posture. When the handshake ends, the system reverses the motion to restore the fingers' posture.

3 System Validation

To validate the active force feedback glove and control algorithm, a simple XR application for a virtual handshake is built. In the scenario, a user wears a head-mounted display (HMD, VIVE Focus Vision, HTC VIVE, Taiwan) and the active force feedback glove equipped with an infrared tracker (VIVE Wrist Tracker, HTC VIVE, Taiwan) (Figure 2). When the XR application launches, an avatar appears in front of the user, bends the elbow, and awaits a handshake. As the user moves their gloved hand near the avatar's hand,

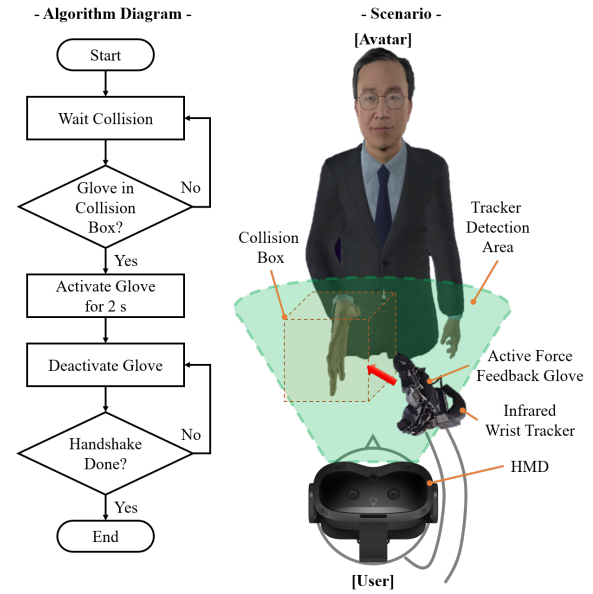


Figure 2: Control algorithm diagram and validation scenario.

the HMD detects the position of the user's hand using the wrist tracker. Once the user's hand enters the collision box around the avatar's hand, the glove is triggered to form a handshake posture, following the avatar's handshake motion. The glove remains in this posture until the avatar releases its hand. Once the handshake process is complete, the glove restores the user's fingers to their initial posture, and the scenario ends.

4 Conclusion and Future Work

In this study, we propose a novel active force feedback glove with an exoskeleton accompanied by a flexible shaft-driven mechanism and its control algorithm, providing active force feedback in response to a variety of virtual dynamic interactions. The system validation demonstrated that the proposed system effectively simulates a virtual handshake in a simple yet effective manner. As a result, users can experience more realistic virtual activities through active force feedback delivered by the proposed system. In future work, we plan to enhance the control algorithm with additional sensors and actuators, such as pressure sensors and micro-vibrators, implementing various virtual dynamic interactions.

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